LAB-5 Software Engineering

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SRN – PES2UG24CS375

SECTION -G

Topic: Campus Shuttle

Output of 3 tools (BEFORE CHANGES):

bandit_report.txt:

```
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle> python -m bandit -r shuttle_system.py
[main]  INFO    running on Python 3.14.3
Run started:2026-10-29 10:18:41.280727+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\shuttle_system.py:29:8
28                      flagged.append(r)
29              except:
30                  pass
31          return flagged

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\shuttle_system.py:35:11
34          drivers = ["Prakash", "Leela", "Imran", "Tom"]
35          return random.choice(drivers)
36

--------------------------------------------------

Code scanned:
        Total lines of code: 50
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 2
Files skipped (0):
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle>
```

pylint_report.txt:

```
Files skipped (0):
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle> python -m pylint shuttle_system.py
************* Module shuttle_system
shuttle_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
shuttle_system.py:11:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:11:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
shuttle_system.py:13:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
shuttle_system.py:16:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:19:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:23:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:29:8: W0702: No exception type(s) specified (bare-except)
shuttle_system.py:33:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:33:0: C0103: Function name "assignDriver" doesn't conform to snake_case naming style (invalid-name)
shuttle_system.py:33:17: W0613: Unused argument 'route_id' (unused-argument)
shuttle_system.py:37:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:38:4: W0603: Using the global statement (global-statement)
shuttle_system.py:39:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
shuttle_system.py:39:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
shuttle_system.py:42:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
shuttle_system.py:42:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
shuttle_system.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)

-------------------------------------------------------------------
Your code has been rated at 6.25/10 (previous run: 6.25/10, +0.00)
```

flake8_report.txt:

```
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle> python -m flake8 shuttle_system.py
shuttle_system.py:11:1: E302 expected 2 blank lines, found 1
shuttle_system.py:13:80: E501 line too long (82 > 79 characters)
shuttle_system.py:16:1: E302 expected 2 blank lines, found 1
shuttle_system.py:19:1: E302 expected 2 blank lines, found 1
shuttle_system.py:23:1: E302 expected 2 blank lines, found 1
shuttle_system.py:29:9: E722 do not use bare 'except'
shuttle_system.py:33:1: E302 expected 2 blank lines, found 1
shuttle_system.py:37:1: E302 expected 2 blank lines, found 1
shuttle_system.py:45:1: E302 expected 2 blank lines, found 1
shuttle_system.py:60:1: E305 expected 2 blank lines after class or function definition, found 1
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle>
```

Output of 3 tools (AFTER CHANGES):

bandit_report.txt:

```
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle> python -m bandit -r shuttle_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.3
Run started:2026-10-29 10:11:19.198596+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 74
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle>
```

pylint_report.txt:

```
[notice] To update, run: python.exe -m pip install --upgrade pip
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle> python -m pylint shuttle_system.py
************* Module shuttle_system
shuttle_system.py:76:0: C0116: Missing function or method docstring (missing-function-docstring)

-------------------------------------------------------------------
Your code has been rated at 9.82/10 (previous run: 9.82/10, +0.00)
```

flake8_report.txt:

```
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle> python -m flake8 shuttle_system.py
PS C:\Users\prith\PES2UG24CS375_SE_LAB5\L5_07_campus_shuttle>
```